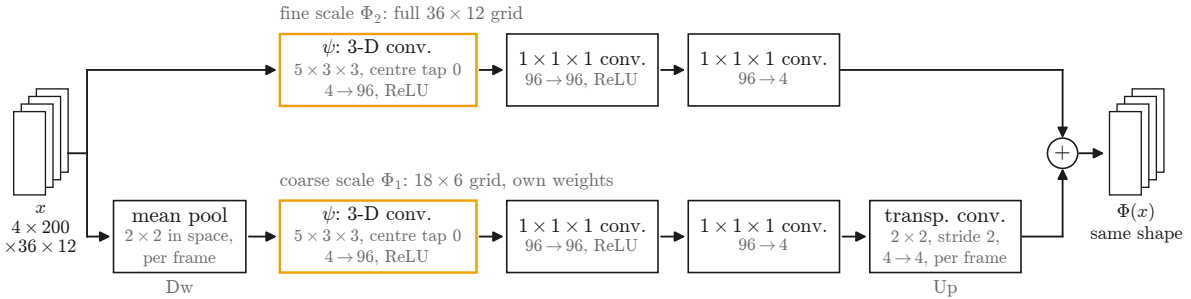
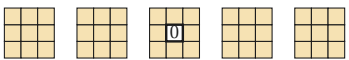
(a) prior $\Phi(x) = \mathrm{Up}(\Phi_1(\mathrm{Dw}(x))) + \Phi_2(x)$




(b) the kernel of $\psi$ (one input/output channel pair)




The tap at the output's own point $(t, i, j)$ is fixed to zero, and the layers after $\psi$ act point by point, so a branch's output at $s$ never reads $x(s)$. Pooling and the stride-2 transposed convolution act per frame, so each fine cell takes one coarse output, which excludes its own pooled cell. Hence $\Phi(x)(s)$ does not depend on $x(s)$ and $\Phi$ cannot learn the identity, which would make $\|x - \Phi(x)\|^2$ vanish.